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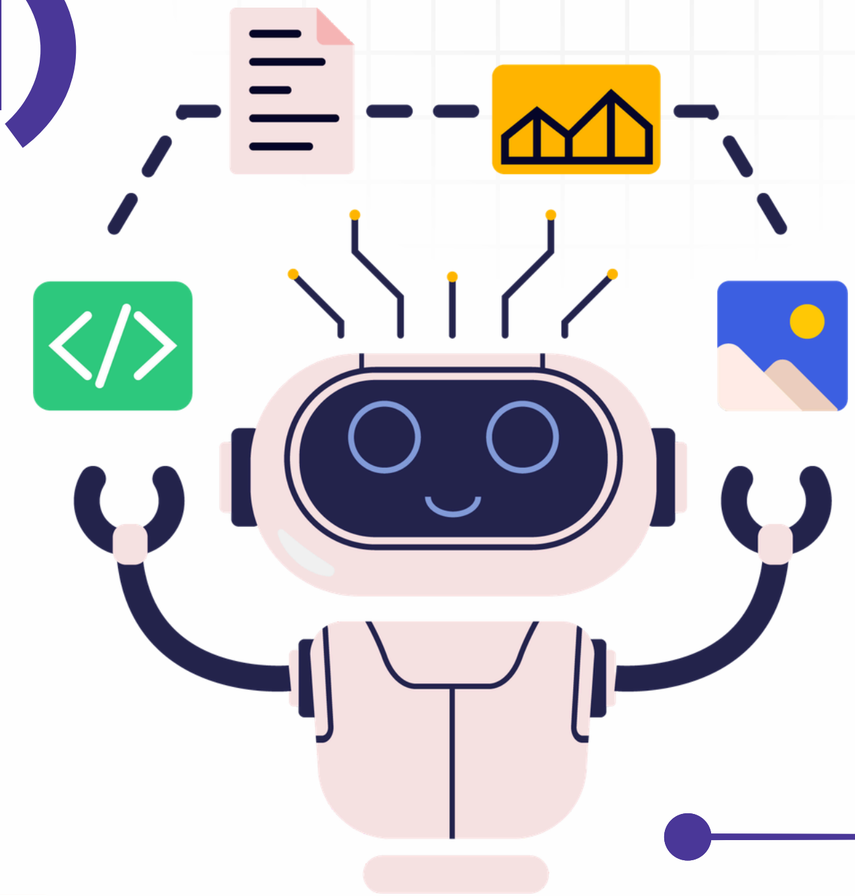
Introduction to AI Ethics

Understanding Ethical AI and Its Challenges



Definition of Artificial Intelligence (AI)

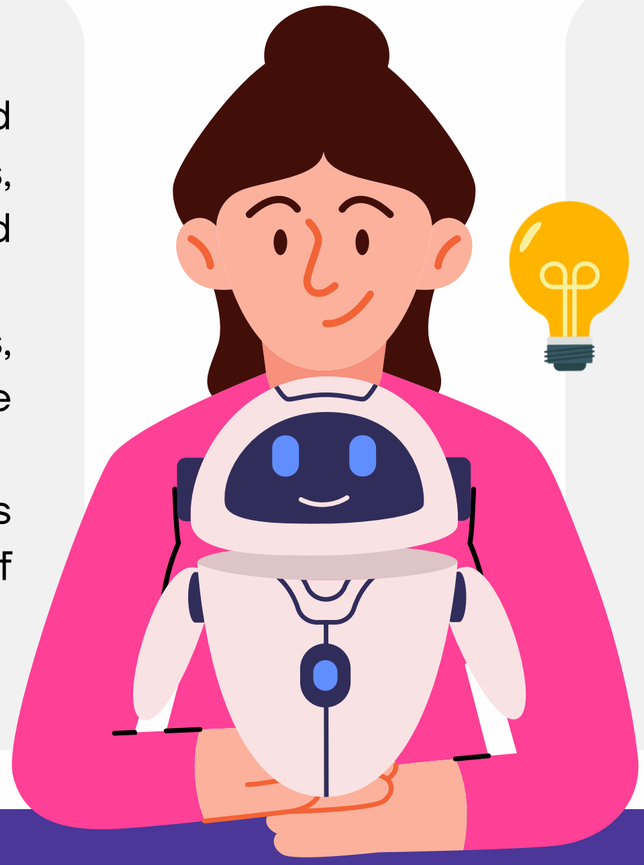
Artificial Intelligence, often abbreviated as AI, is technology that supports computers in acting and thinking like humans. Since its development more than 70 years ago, AI has continued to evolve. Today, AI can simulate human communication and create various forms of creative content, such as text, images, video, audio, and more.



Impact of AI Use on Human Life

POSITIVE IMPACTS

- Increases efficiency, automation, and productivity across various sectors, including finance, healthcare, and data management.
- Makes telecommunications, information, and education more accessible worldwide.
- Helps humans identify social problems through analysis of large amounts of data.



NEGATIVE IMPACTS

- Replaces human roles across various sectors, causing some jobs to potentially disappear.
- The use and development of AI can create privacy and security issues for wider society.
- Excessive use of AI can create dependency and reduce humans' ability to think rationally.

CHALLENGES IN AI USE 01

Algorithmic Bias

Algorithmic bias occurs in AI systems when predictions or decisions are influenced by biases in the training data or the model's design. This issue often leads to unfair outcomes that do not consider all groups equally. For example, AI-driven recruitment systems may exhibit discrimination based on gender, race, or socioeconomic background.





CHALLENGES IN AI USE 02

Data Privacy

AI models require large amounts of data to function accurately. However, this data is often collected without individuals' full consent or awareness. Once gathered, sensitive information—such as personal financial data, demographic details, and biometric data (facial features and voice recordings)—can be misused or managed beyond the owner's control.

CHALLENGES IN AI USE 03

Data Security

If not properly secured, the data used by AI becomes vulnerable to cyberattacks. Large-scale data breaches can lead to the loss of critical personal information. Notably, data breach incidents have steadily increased over the past five years.



What is AI Ethics and Why Does It Matter?

AI ethics is the multidisciplinary study and practice of optimizing the positive and minimizing the negative impacts of AI technology. AI ethics considers the needs and interests of society, users, and AI developers in addressing challenges related to bias, privacy, and data security.



Principles of AI Ethics

RESPECT FOR INDIVIDUALS

Developers must request permission before using data belonging to data owners and comply with applicable regulations. Developers must also ensure that data is used appropriately for all users.





AI algorithms uphold the principles of equality without considering gender, race, socioeconomic status, or other factors that could lead to discrimination.

JUSTICE

TRANSPARENCY

Developers should provide information about how user data is processed in an honest and easily understandable manner to build trust.





AI Ethics Violation Case Study

Case	A job recruitment algorithm only selects male applicants for certain positions.
AI Ethics Principle Violated	Fairness
Reason	The system demonstrates bias because the algorithm only considers male applicants. This indicates discrimination against female applicants who may also be qualified.
Solution	The company should review its data and improve the system to prevent similar problems. The AI algorithm must not consider gender as a selection factor.



Closing

Technology is advancing rapidly, and its applications in daily life continue to diversify across many sectors. Therefore, AI ethics is essential to ensure responsible innovation. Let's continue exploring the world of AI ethics to expand our knowledge!